

Stage 2 — The location law and its mechanism

Where the Shortcut Sits: progress report 2

In one sentence. When only the artifact’s overlap with the region of interest (ROI) is varied, ROI masking helps while the artifact lies outside the ROI and loses its benefit — in 7 of 11 sweeps it hurts — once the artifact overlaps it; the reversal comes from the artifact *surviving* the mask and becoming what the model relies on, not from occlusion; and it recurs in thyroid and ovarian ultrasound, capsule endoscopy and chest radiography, with one principled exception (an encoder that ignores an artifact outside the lungs).

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma (dermoscopy), malignant nodule (thyroid), tumour with a solid component (ovary), erosion (capsule), pneumothorax (chest radiograph).
$A \in \{0, 1\}$	artifact present: a synthetic artifact pasted at a controlled position: ruler (dermoscopy), sonographer caliper (thyroid, ovary), debris blob (capsule endoscopy), tube (chest radiograph).
ROI	region of interest: lesion, nodule, tumour, annotated lesion box, lungs; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	the <i>location interaction</i> $[\text{mask} - \text{ERM}]_{r=0} - [\text{mask} - \text{ERM}]_{r=1}$ of reversed-test AUROC on the same images.
CI	95% crossed seed×image bootstrap interval (Stage 1); every interval in this report uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.

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1 Where this stage starts

Stage 1 re-derived the pilot and fixed the protocol: trap environments, frozen foundation-model features, linear heads chosen on clean validation data, grouped splits, pre-registration and the crossed bootstrap. It showed that the pilot’s central observation replicates: in the controlled dermoscopy sweep, masking changes reversed-test AUROC from $+0.184$ [$+0.117, +0.250$] at 0% overlap to -0.129 [$-0.195, -0.064$] at 100% overlap. Stage 1 deliberately did not interpret the pilot’s mechanism experiments. This stage asks the two questions that follow: *why* does masking reverse, and *how general* is the reversal — does it depend on dermoscopy, on the ruler, on DINOv2?

2 Questions

- Q2a** Is the benefit of masking a function of the artifact’s overlap r with the ROI, falling from $r = 0$ to $r = 1$? We call a positive location interaction $[\text{mask} - \text{ERM}]_{r=0} - [\text{mask} - \text{ERM}]_{r=1}$ on the reversed test the *location law*.
- Q2b** Which mechanism produces the fall? Three candidates were stated in advance: *occlusion* (masking removes useful context whatever the artifact does), *lost disease signal* (the background carries diagnostic information), and *retention with amplified reliance* (the part of the artifact inside the ROI survives masking, its competitors are removed, and the head relies on it more).
- Q2c** Does the law hold beyond one encoder, one artifact rendering and one modality — and where does it fail?

2.1 Why these experiments, and what was rejected

- *An occlusion control instead of an argument from intuition.* If masking harmed because it throws away context, it would harm even when the artifact carries no label information (cf. Fong and Vedaldi, 2019). The control keeps the ruler at the same positions but makes it independent of the label.
- *Dilation as a dose of retention.* Growing the mask by a margin keeps more of the ruler visible without changing the lesion. If retention drives the harm, harm should grow with retention when retention is incomplete (50% overlap) and stay flat when the ruler is already fully retained (100% overlap) — even though dilation then *lowers* the ruler’s share of the visible pixels.
- *A same-head counterfactual instead of saliency maps.* Saliency maps are qualitative and unstable for frozen encoders. Instead, the ruler is inserted into and removed from the *same* test image and the change in each trained head’s predicted probability, $|\Delta p|$, is read: a direct, per-image measure of reliance on the artifact.
- *Lesion size as a natural dose.* The same ruler covers a larger share of a small lesion. If what matters is how dominant the retained artifact is among the remaining evidence, harm should be largest for small lesions. No threshold ratio was pre-specified, because the pilot showed none.
- *Other modalities with their own artifact.* A caliper drawn like a sonographer’s, a debris blob drawn like capsule contamination and a tube drawn like a drain, each pasted at controlled overlap on real images of that modality with that modality’s own ROI masks. This tests whether the law is a property of dermoscopy or of masking.
- *Rejected:* measuring reliance through head weights (they are not comparable between masked and unmasked feature spaces), and a single “average” overlap (it would hide the sign change).

3 Methods

Controlled dermoscopy sweep As in Stage 1: ISIC 2018 pilot cohort, a 65×19 px ruler at a frozen position per image and overlap $r \in \{0, 0.25, 0.5, 0.75, 1\}$; DINOv2 ViT-B/14 at 518 px (Oquab et al., 2024) (CLS features), logistic heads, seeds 42, 123, 456; primary outcome reversed-test AUROC.

Arms ERM; ROI masking; oracle inpainting of the ruler; group-balanced reweighting; DFR (Kirichenko et al., 2023); paired LEACE (Belrose et al., 2023); GroupDRO (Sagawa et al., 2020). All arms share features and seeds; only the input or the head changes.

Occlusion control The ruler is pasted at the same positions in 50% of images independently of the label; AUROC is measured on a 50/50 test.

Mask dilation The lesion mask is dilated by 0, 10, 25 or 50 px before masking. *Retention* is the share of ruler pixels that remain visible; *ruler share* is the ruler’s share of all visible pixels.

Same-head counterfactual For each artifact-bearing image of the reversed test and each trained head, the image is scored with and without the pasted artifact (the masked head sees the masked version of both); $|\Delta p|$ is the absolute change in predicted probability. The mask – ERM difference of mean $|\Delta p|$ is bootstrapped with seeds and images crossed.

Lesion-size strata Test images are split into tertiles of lesion area; the median ruler-to-lesion area ratio is reported per tertile.

Replications DINOv2 at 224 px, DermLIP at 224 px (Yan et al., 2025), and a ruler whose colour, tick pattern and size are randomised per image (“stress”).

Other modalities Each sweep uses only images of that cohort in which the real artifact is absent, so that the pasted artifact is the only one:

- **Thyroid ultrasound** (TN3K images with TNCD malignancy labels and manual nodule masks (Gong et al., 2021); 1,663 caliper-free images): a synthetic sonographer caliper (dotted line with “+” ends) in the same 65×19 box as the ruler; DINOv2 and MedSigLIP-448. Split: official TN3K test images as test; the rest 80/20 by leakage group.
- **Ovarian ultrasound** (MMOTU OTU_2d (Zhao et al., 2022); benign cystic lesions versus tumours with a solid component; 343 caliper-free images, 64 in test): the same caliper; DINOv2. Small, so wide intervals were expected.
- **Capsule endoscopy** (SEE-AI (Yokote et al., 2023); erosion versus polyp-like lesions; ROI = the union of the expert lesion boxes; 796 low-contamination frames, split 60/20/20 by frame-block group): a textured yellow-green debris blob with bubbles, 48×48 px; DINOv2 and MedSigLIP.
- **Chest radiography** (NIH ChestX-ray14 pneumothorax (Wang et al., 2017); lung masks from a PSPNet (Cohen et al., 2022); patient-grouped split): a synthetic tube; RAD-DINO @518 (Pérez-García et al., 2025) (a chest-radiograph foundation model) and DINOv2.

Figure 2 shows three of these sweeps on real images.

Statistics Every interval is a 95% percentile interval from 10,000 crossed bootstrap replicates. “Significant” means the 95% interval excludes zero; it is not a multiplicity-adjusted test.

4 Experiments

Table 1 lists each experiment of this stage with its registration and support criterion. The dermoscopy analyses (E1–E7) were specified in the pilot specification (2699d51, 2026-09-26 15:47); the other-modality sweeps were registered with their cohorts before any model was fitted. The DINOv2 chest-tube sweep was added after the RAD-DINO result and is labelled post hoc.

Table 1: Experiments of this stage: registration status, the registration document (docs/PREREGISTRATION_<name>.md or the replication specification), the commit that first contains it, and what was stated in advance to count as support. Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Controlled dermoscopy sweep (E1)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 interaction CI > 0; mask-ERM CI > 0 at 0% and < 0 at 100%
Replications: DINOv2@224, DermLIP, randomised ruler	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 same signs and verdicts as E1
Occlusion control (E4)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 no harm on the 50/50 test at any overlap
Mask dilation (E5)	supported registered	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 harm grows with retention at 50%; flat at 100%
Same-head counterfactual $ \Delta p $ (E6, T3)	supported registered	REVIEW2 (T3)	29cd0da, 18:56	2026-09-27 mask > ERM at in-ROI overlap, CI > 0
Lesion-size tertiles (E7)	partly supported pilot analysis, replicated	REPLICATION_SPEC	2699d51, 15:47	2026-09-26 harm largest in small lesions
Thyroid caliper sweep	supported registered	THYROID_TRAPS	ea57661, 22:46	2026-09-26 interaction CI > 0
Ovarian caliper sweep	supported registered	OVARY_TRAPS	2517600, 07:29	2026-09-27 interaction CI > 0
Capsule debris sweep	supported registered	CAPSULE_TRAPS	fcfb8d9, 00:40	2026-09-27 interaction CI > 0
Chest tube sweep (RAD-DINO)	supported registered	CXR_DEVICE_TRAPS	4d1a924, 18:07	2026-09-26 interaction CI > 0
Chest tube sweep (DINOv2)	not supported post hoc	—	—	descriptive only
Crossed-bootstrap intervals for every sweep (A1)	supported registered	FINAL (A1)	c45d482, 00:35	2026-09-28 a claim is stated only if its crossed CI excludes 0

5 Results

5.1 The dermoscopy sweep: every arm, every overlap

Masking changes the reversed-test AUROC by +0.184 [+0.117, +0.250] when the ruler lies outside the lesion, by −0.097 [−0.164, −0.030] at 50% overlap and by −0.129 [−0.195, −0.064] at full overlap (Table 2); the location interaction is +0.313 [+0.258, +0.370]. Masking therefore lowers it significantly at 50, 75 and 100% overlap. Every other arm helps at every overlap, and their benefit *grows* with overlap — the opposite of masking: balancing’s location interaction is −0.143 [−0.171, −0.119], inpainting’s −0.103 [−0.127, −0.080]. They target the artifact, not the background, and the ERM shortcut they remove is larger when the ruler sits on the lesion. On the correlated test the masked model at full overlap scores 0.961 against 0.937 for ERM: the masked model exploits the ruler more.

Table 2: Controlled dermoscopy sweep (ISIC 2018, DINOv2 ViT-B/14 @518, three seeds): change in reversed-test AUROC versus ERM, crossed CIs.

	0% overlap	50%	100%
ERM (reversed AUROC)	0.665	0.573	0.515
ROI masking – ERM	+0.184 [+0.117, +0.250]	−0.097 [−0.164, −0.030]	−0.129 [−0.195, −0.064]
oracle inpainting – ERM	+0.141 [+0.109, +0.175]	+0.211 [+0.175, +0.249]	+0.243 [+0.205, +0.284]
group-balanced – ERM	+0.137 [+0.099, +0.179]	+0.219 [+0.177, +0.265]	+0.281 [+0.236, +0.329]
DFR – ERM	+0.124 [+0.075, +0.179]	+0.221 [+0.166, +0.281]	+0.279 [+0.220, +0.339]
paired LEACE – ERM	+0.167 [+0.126, +0.211]	+0.237 [+0.197, +0.280]	+0.294 [+0.249, +0.339]
GroupDRO – ERM	+0.080 [+0.044, +0.118]	+0.152 [+0.112, +0.193]	+0.197 [+0.155, +0.240]

Replications The sweep reverses with DINOv2 at 224 px (interaction +0.456 [+0.390, +0.525]), with DermLIP (+0.424 [+0.335, +0.517]) and with a randomised ruler (+0.302 [+0.248, +0.357]) (Table 3).

Table 3: Replications of the controlled sweep (reversed-test AUROC, crossed CIs).

Backbone / artifact	mask – ERM at 0%	at 100%	Location interaction
DINOv2 @224	+0.361 [+0.275, +0.444]	−0.095 [−0.156, −0.035]	+0.456 [+0.390, +0.525]
DermLIP @224	+0.161 [+0.102, +0.222]	−0.263 [−0.348, −0.180]	+0.424 [+0.335, +0.517]
DINOv2 @518, randomised ruler	+0.146 [+0.082, +0.211]	−0.155 [−0.228, −0.084]	+0.302 [+0.248, +0.357]

5.2 Mechanism

Occlusion does not explain the harm With the ruler present but uninformative, masking changes it significantly at none of the five overlaps (every CI includes zero) (Table 4); inpainting, which removes only the ruler, changes nothing either. Masking by itself does not cost the reversed-test AUROC it loses in the sweep.

Table 4: Occlusion control: ruler present but uncorrelated with the label (50/50); AUROC change on the 50/50 test.

Overlap	mask – ERM	inpaint – ERM
0%	+0.036 [−0.016, +0.086]	−0.001 [−0.005, +0.003]
25%	+0.022 [−0.030, +0.073]	−0.010 [−0.034, +0.002]
50%	+0.041 [−0.010, +0.090]	−0.002 [−0.007, +0.003]
75%	+0.037 [−0.013, +0.086]	−0.003 [−0.008, +0.001]
100%	+0.039 [−0.010, +0.091]	−0.004 [−0.010, +0.002]

Harm follows the visibility of the artifact At 50% overlap, dilating the mask from 0 to 25 px raises retention from 0.500 to 0.946 and moves the masking effect from −0.097 [−0.164, −0.031] to −0.153 [−0.221, −0.089]. At 100% overlap the ruler is already retained (0.999) and the effect stays flat (−0.129 [−0.195, −0.066] without and −0.138 [−0.200, −0.082] with a 50 px margin), although dilation *lowers* the ruler’s share of the visible pixels threefold (0.055 to 0.018). Harm therefore tracks whether the ruler is visible, not how much of the image it occupies once visible (Table 5). This also rules out “lost context” as the sole cause: dilation *restores* context, yet harm grows at 50% overlap. The intervals of neighbouring margins overlap; the pattern, not each step, is the finding.

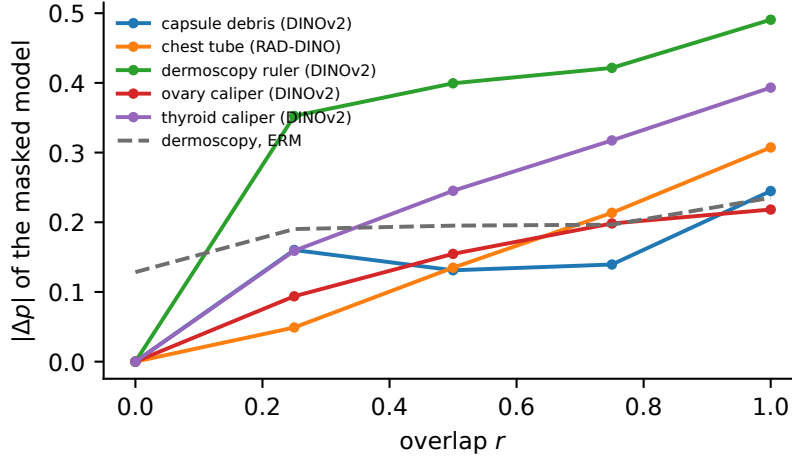
Table 5: Mask dilation: restoring context around the lesion restores ruler pixels (retention) and changes harm.

Overlap	Margin (px)	Ruler retained	Ruler share of visible pixels	mask – ERM (rev)
50%	0	0.500	0.028	−0.097 [−0.164, −0.031]
	10	0.725	0.029	−0.138 [−0.203, −0.074]
	25	0.946	0.026	−0.153 [−0.221, −0.089]
	50	1.000	0.018	−0.142 [−0.206, −0.083]
75%	0	0.750	0.042	−0.111 [−0.181, −0.044]
	10	0.940	0.037	−0.152 [−0.217, −0.091]
	25	1.000	0.028	−0.160 [−0.226, −0.096]
	50	1.000	0.018	−0.140 [−0.201, −0.079]
100%	0	0.999	0.055	−0.129 [−0.195, −0.066]
	10	1.000	0.040	−0.144 [−0.206, −0.082]
	25	1.000	0.028	−0.144 [−0.209, −0.082]
	50	1.000	0.018	−0.138 [−0.200, −0.082]

The masked model relies more on the retained artifact On the same images, inserting the ruler at full overlap changes the masked head’s prediction by 0.491 on average and the ERM head’s by 0.235 — about 2.1 times as much; the paired difference is +0.255 [+0.221, +0.287] (Table 6, Figure 1). At 0% overlap the masked head cannot see the ruler (0.000) whereas ERM still reacts to it (0.128). This is the mechanism in one measurement: masking removes the competing evidence, the retained artifact becomes the most informative remaining feature, and the head leans on it. Across the 5 sweeps with counterfactual scores the masked head is significantly more sensitive at full overlap in 3: ovarian calipers (0.218 versus 0.032; +0.188 [+0.133, +0.250]) and capsule debris (0.245 versus 0.057; +0.186 [+0.130, +0.260]) besides the ruler. For thyroid calipers the two heads are equally sensitive (+0.025 [−0.016, +0.065]) — ERM already relies heavily on a caliper inside the nodule — and with RAD-DINO the masked head is *less* sensitive to an in-lung tube (−0.127 [−0.136, −0.118]).

Table 6: Same-head counterfactual sensitivity to inserting the artifact (reversed test, artifact-bearing images), crossed CIs of the paired difference.

Sweep	Overlap	$ \Delta p $ ERM	$ \Delta p $ mask	mask – ERM
dermoscopy ruler (DINOv2)	0%	0.128	0.000	−0.128 [−0.136, −0.121]
	25%	0.190	0.353	+0.162 [+0.136, +0.188]
	50%	0.195	0.399	+0.204 [+0.177, +0.230]
	75%	0.197	0.422	+0.225 [+0.195, +0.255]
	100%	0.235	0.491	+0.255 [+0.221, +0.287]
thyroid caliper (DINOv2)	0%	0.285	0.000	−0.285 [−0.317, −0.255]
	25%	0.353	0.159	−0.195 [−0.271, −0.131]
	50%	0.351	0.245	−0.106 [−0.176, −0.046]
	75%	0.359	0.317	−0.041 [−0.078, −0.005]
	100%	0.369	0.393	+0.025 [−0.016, +0.065]
ovary caliper (DINOv2)	0%	0.016	0.000	−0.016 [−0.020, −0.013]
	25%	0.027	0.094	+0.067 [+0.040, +0.098]
	50%	0.027	0.155	+0.129 [+0.090, +0.174]
	75%	0.031	0.198	+0.168 [+0.125, +0.217]
	100%	0.032	0.218	+0.188 [+0.133, +0.250]
capsule debris (DINOv2)	0%	0.033	0.000	−0.033 [−0.041, −0.025]
	25%	0.079	0.160	+0.081 [+0.001, +0.149]
	50%	0.053	0.131	+0.077 [+0.031, +0.140]
	75%	0.058	0.139	+0.081 [+0.058, +0.109]
	100%	0.057	0.245	+0.186 [+0.130, +0.260]
chest tube (RAD-DINO)	0%	0.005	0.000	−0.005 [−0.006, −0.005]
	25%	0.095	0.049	−0.046 [−0.051, −0.041]
	50%	0.260	0.135	−0.126 [−0.134, −0.117]
	75%	0.355	0.214	−0.141 [−0.151, −0.130]
	100%	0.434	0.307	−0.127 [−0.136, −0.118]

Figure 1: Mean same-head sensitivity $|\Delta p|$ to inserting the artifact: masked model per sweep (solid) and ERM for the dermoscopy sweep (dashed), by overlap.

Lesion size Table 7. The ruler covers a median 0.012 of a large lesion’s area, 0.031 of a medium and 0.106 of a small one. The harm at full overlap is significant for small (−0.364 [−0.509, −0.217]) and medium lesions (−0.158 [−0.295, −0.027]) and not for large ones (+0.063 [−0.015, +0.144], which does not differ significantly from zero); at 50% overlap it is significant only for small lesions (−0.326 [−0.471, −0.176]; medium −0.129 [−0.283, +0.027]). What holds is an ordering — harm is

largest where the artifact is large relative to the lesion — with wide intervals, because each tertile has few melanomas. No fixed ratio threshold is claimed.

Table 7: Lesion-size strata of the controlled sweep (reversed-test AUROC, crossed CIs).

Lesion tertile	Ruler/lesion area (median)	Melanomas in test	mask – ERM at 0%	50%	100%
large	0.012	36	+0.249 [+0.155, +0.346]	+0.087 [+0.006, +0.172]	+0.063 [−0.015, +0.144]
medium	0.031	21	+0.215 [+0.033, +0.391]	−0.129 [−0.283, +0.027]	−0.158 [−0.295, −0.027]
small	0.106	16	+0.150 [+0.036, +0.273]	−0.326 [−0.471, −0.176]	−0.364 [−0.509, −0.217]

Synthesis Occlusion alone does not harm; restoring context does not remove the harm; harm follows whether the artifact is visible after masking, and the masked model’s reliance on a visible artifact is larger than ERM’s. The mechanism is *retention with amplified reliance*: masking deletes the out-of-ROI evidence — including other cues that competed with the artifact — and leaves the in-ROI artifact as a larger share of what predicts the label in training, so the head weights it more heavily.

5.3 Other modalities

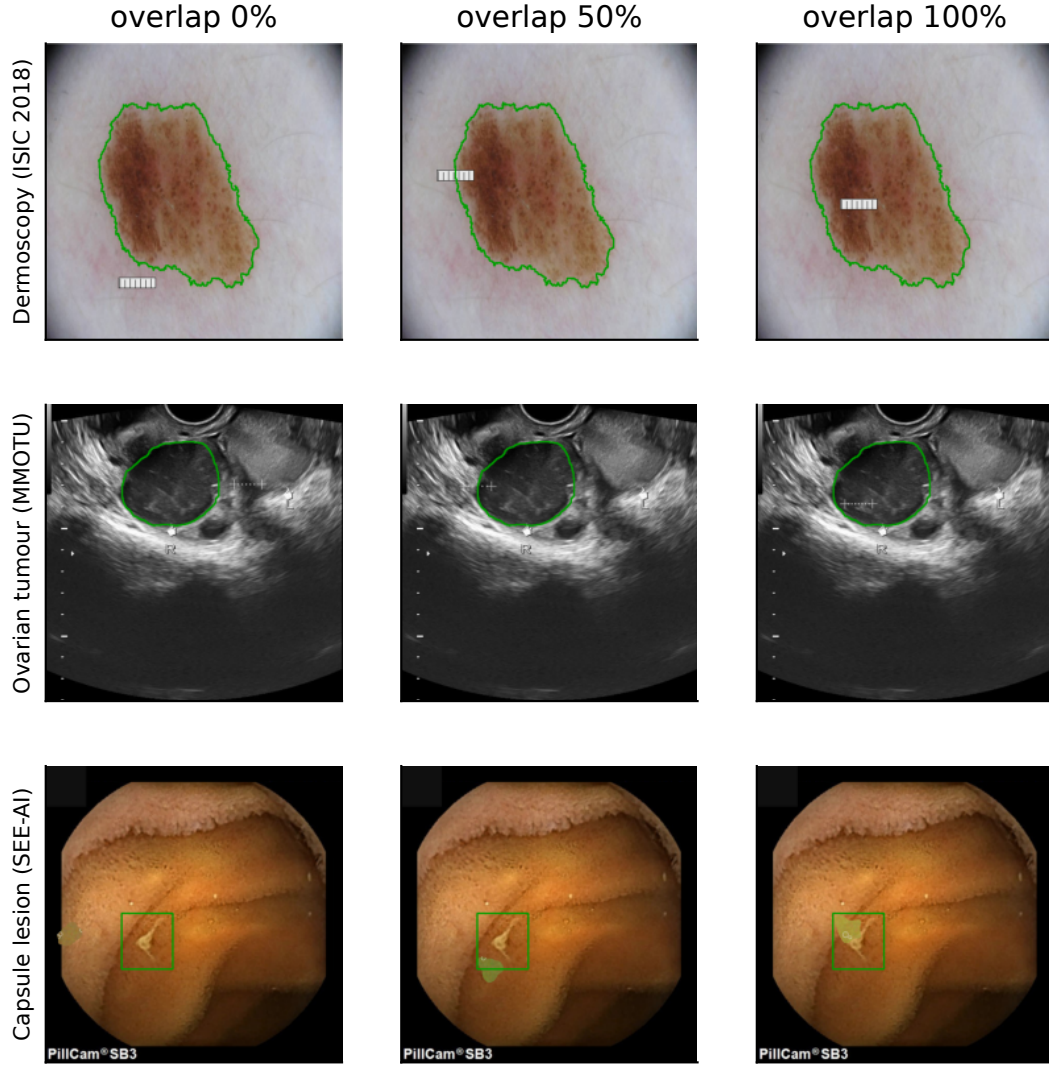


Figure 2: Real images with the synthetic artifact at 0, 50 and 100% overlap (green: ROI; the capsule ROI is the annotated bounding box). Top: ISIC 2018 (CC BY-NC 4.0); middle: MMOTU ovarian ultrasound (CC BY 4.0); bottom: SEE-AI capsule endoscopy (CC BY 4.0). Thyroid (TN3K) and NIH images are not shown because their redistribution terms are unclear.

Table 8 and Figure 3 repeat the sweep in four further modalities; Table 9 gives every baseline arm. Across all 11 controlled sweeps (four dermoscopy, seven others) the location interaction is positive with an interval above zero in 10 and negative in 1. Whether the falling benefit turns into harm at full overlap depends on the setting: masking harms significantly at full overlap in 7 sweeps, still helps in 2 and makes no significant difference in 2.

Table 8: Controlled sweeps in the other modalities (synthetic caliper, debris, tube pasted at controlled overlap; reversed-test AUROC, crossed CIs).

Sweep	mask – ERM at 0%	at 100%	Location interaction
Thyroid caliper, DINOv2	+0.402 [+0.315, +0.489]	−0.030 [−0.105, +0.042]	+0.433 [+0.362, +0.507]
Thyroid caliper, MedSigLIP	+0.514 [+0.430, +0.600]	+0.044 [−0.011, +0.104]	+0.470 [+0.388, +0.556]
Ovarian caliper, DINOv2	+0.041 [−0.096, +0.175]	−0.197 [−0.357, −0.035]	+0.238 [+0.144, +0.340]
Capsule debris, DINOv2	+0.116 [+0.052, +0.191]	−0.112 [−0.209, −0.019]	+0.228 [+0.144, +0.324]
Capsule debris, MedSigLIP	+0.374 [+0.248, +0.509]	+0.216 [+0.118, +0.317]	+0.158 [+0.012, +0.314]
Chest tube, RAD-DINO	−0.002 [−0.010, +0.005]	+0.111 [+0.095, +0.127]	−0.113 [−0.129, −0.097]
Chest tube, DINOv2	+0.254 [+0.230, +0.278]	−0.083 [−0.112, −0.057]	+0.337 [+0.312, +0.364]

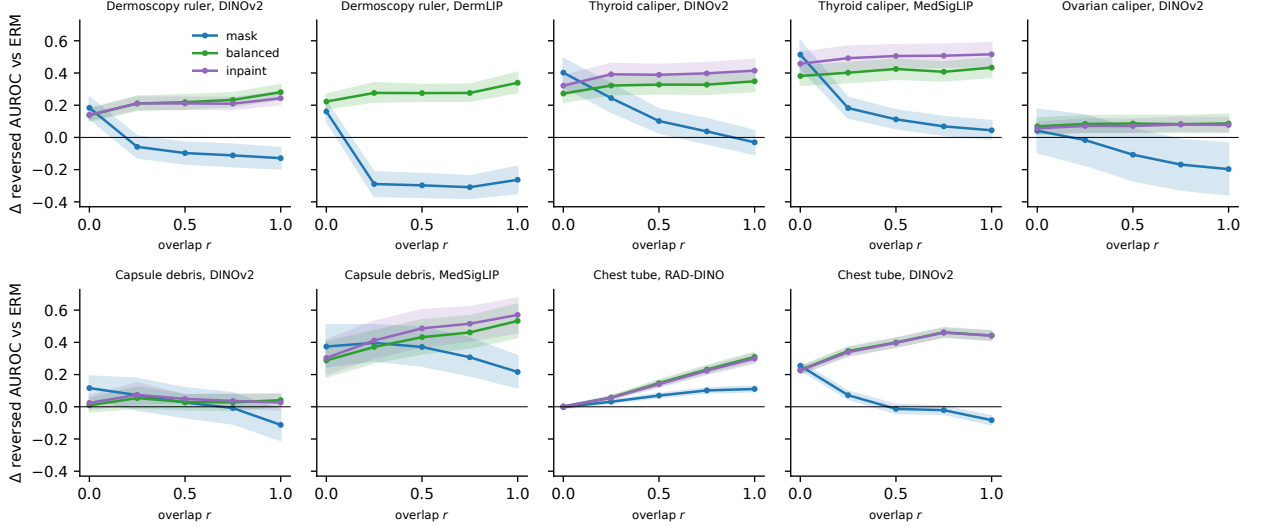


Figure 3: Change in reversed-test AUROC versus ERM as a function of overlap, for masking (blue), group balancing (green) and oracle inpainting (purple), in every controlled sweep (shaded: crossed 95% CI). Masking falls with overlap in every panel except the RAD-DINO chest tube.

- **Thyroid calipers.** Masking helps strongly with the caliper outside the nodule (DINOv2 +0.402 [+0.315, +0.489]; MedSigLIP +0.514 [+0.430, +0.600]) and the benefit vanishes at full overlap (−0.030 [−0.105, +0.042]; +0.044 [−0.011, +0.104]): the law holds as a loss of benefit, not as harm. Nodules are large relative to the caliper, and the ultrasound background carries little disease information, so masking costs little besides leaving the caliper.
- **Ovarian calipers.** Masking harms at full overlap (−0.197 [−0.357, −0.035]) and its benefit outside the tumour is not significant (+0.041 [−0.096, +0.175]): MMOTU tumours fill much of the frame, so little background is removed, and the test set is small (64 images).
- **Capsule debris.** With DINOv2 the sweep reverses (+0.116 [+0.052, +0.191] to −0.112 [−0.209, −0.019]). With MedSigLIP masking still helps at full overlap (+0.216 [+0.118, +0.317]) but less than outside (+0.374 [+0.248, +0.509]; interaction +0.158 [+0.012, +0.314]): the box ROI contains much normal mucosa, so masking removes little of the lesion’s context.
- **Chest tube — the boundary.** With RAD-DINO the ERM model ignores a tube outside the lungs (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove at $r = 0$; at full overlap ERM collapses (0.614) and the masked model less so (0.725). The interaction is negative. With DINOv2, which does use the extra-pulmonary tube (reversed 0.629 versus clean 0.859), the law holds (+0.337 [+0.312, +0.364]). The law’s premise is that the encoder uses the out-of-ROI artifact; when it does not, masking cannot help outside and the law does not apply.
- **The artifact-targeting arms** (inpainting, balancing, DFR, paired LEACE) help at both overlaps in every modality except where the intervals are wide (ovary, DINOv2 capsule; Table 9).

None of them shows the positive location interaction of masking; most show a small negative one, because ERM relies more on an artifact that sits on the lesion and there is more for them to remove.

Table 9: All baseline arms in the controlled sweeps of the other modalities: ERM’s clean and reversed AUROC and each arm’s change in reversed-test AUROC versus ERM at no and at full overlap (crossed CIs). The last column is the arm’s own location interaction ($[\cdot]_{r=0} - [\cdot]_{r=1}$); an arm that does not depend on where the artifact sits has an interaction near zero.

Sweep	Arm	$r = 0$	$r = 1$	Location interaction
Thyroid caliper, DI-NOv2	ERM (AUROC: clean; reversed)	0.666; 0.417	0.643; 0.325	—
	masking — ERM	+0.402 [+0.315, +0.489]	−0.030 [−0.105, +0.042]	+0.433 [+0.362, +0.507]
	oracle inpainting — ERM	+0.321 [+0.261, +0.383]	+0.415 [+0.348, +0.484]	−0.093 [−0.124, −0.065]
	group-balanced — ERM	+0.273 [+0.220, +0.328]	+0.349 [+0.286, +0.415]	−0.076 [−0.106, −0.048]
	DFR — ERM	+0.299 [+0.228, +0.374]	+0.384 [+0.298, +0.472]	−0.085 [−0.123, −0.050]
	paired LEACE — ERM	+0.324 [+0.264, +0.386]	+0.420 [+0.353, +0.488]	−0.096 [−0.129, −0.066]
Thyroid caliper, MedSigLIP	ERM (AUROC: clean; reversed)	0.742; 0.320	0.723; 0.260	—
	masking — ERM	+0.514 [+0.430, +0.600]	+0.044 [−0.011, +0.104]	+0.470 [+0.388, +0.556]
	oracle inpainting — ERM	+0.457 [+0.388, +0.527]	+0.516 [+0.442, +0.589]	−0.059 [−0.089, −0.031]
	group-balanced — ERM	+0.381 [+0.324, +0.442]	+0.433 [+0.373, +0.494]	−0.052 [−0.084, −0.020]
	DFR — ERM	+0.437 [+0.370, +0.505]	+0.484 [+0.409, +0.561]	−0.047 [−0.086, −0.008]
	paired LEACE — ERM	+0.460 [+0.389, +0.531]	+0.517 [+0.442, +0.587]	−0.057 [−0.089, −0.027]
Ovarian caliper, DI-NOv2	ERM (AUROC: clean; reversed)	0.668; 0.656	0.677; 0.635	—
	masking — ERM	+0.041 [−0.096, +0.175]	−0.197 [−0.357, −0.035]	+0.238 [+0.144, +0.340]
	oracle inpainting — ERM	+0.057 [+0.025, +0.092]	+0.077 [+0.034, +0.122]	−0.021 [−0.047, +0.004]
	group-balanced — ERM	+0.069 [+0.020, +0.121]	+0.087 [+0.034, +0.143]	−0.018 [−0.048, +0.009]
	DFR — ERM	+0.013 [−0.079, +0.102]	+0.047 [−0.049, +0.142]	−0.034 [−0.077, +0.005]
	paired LEACE — ERM	+0.056 [+0.023, +0.093]	+0.073 [+0.031, +0.119]	−0.017 [−0.044, +0.011]
Capsule debris, DI-NOv2	ERM (AUROC: clean; reversed)	0.853; 0.822	0.849; 0.797	—
	masking — ERM	+0.116 [+0.052, +0.191]	−0.112 [−0.209, −0.019]	+0.228 [+0.144, +0.324]
	oracle inpainting — ERM	+0.024 [−0.016, +0.069]	+0.026 [−0.020, +0.074]	−0.002 [−0.026, +0.024]
	group-balanced — ERM	+0.009 [−0.032, +0.053]	+0.041 [+0.004, +0.081]	−0.032 [−0.073, +0.001]
	DFR — ERM	+0.030 [−0.034, +0.090]	+0.049 [−0.016, +0.114]	−0.019 [−0.041, +0.001]
	paired LEACE — ERM	+0.035 [+0.014, +0.061]	+0.059 [+0.030, +0.094]	−0.023 [−0.043, −0.006]
Capsule debris, MedSigLIP	ERM (AUROC: clean; reversed)	0.902; 0.586	0.887; 0.302	—
	masking — ERM	+0.374 [+0.248, +0.509]	+0.216 [+0.118, +0.317]	+0.158 [+0.012, +0.314]
	oracle inpainting — ERM	+0.303 [+0.201, +0.414]	+0.570 [+0.459, +0.675]	−0.267 [−0.388, −0.145]
	group-balanced — ERM	+0.288 [+0.185, +0.402]	+0.533 [+0.430, +0.636]	−0.245 [−0.359, −0.130]
	DFR — ERM	+0.294 [+0.186, +0.416]	+0.575 [+0.453, +0.695]	−0.281 [−0.398, −0.160]
	paired LEACE — ERM	+0.319 [+0.212, +0.433]	+0.608 [+0.499, +0.711]	−0.289 [−0.405, −0.171]
Chest tube, RAD-DINO	ERM (AUROC: clean; reversed)	0.915; 0.913	0.907; 0.614	—
	masking — ERM	−0.002 [−0.010, +0.005]	+0.111 [+0.095, +0.127]	−0.113 [−0.129, −0.097]
	oracle inpainting — ERM	+0.003 [+0.002, +0.004]	+0.299 [+0.271, +0.329]	−0.297 [−0.326, −0.267]
	group-balanced — ERM	+0.000 [−0.004, +0.004]	+0.309 [+0.280, +0.339]	−0.309 [−0.338, −0.281]
	DFR — ERM	−0.014 [−0.022, −0.007]	+0.280 [+0.248, +0.313]	−0.294 [−0.328, −0.262]
	paired LEACE — ERM	+0.003 [+0.002, +0.004]	+0.287 [+0.259, +0.316]	−0.284 [−0.314, −0.255]
Chest tube, DI-NOv2	ERM (AUROC: clean; reversed)	0.859; 0.629	0.852; 0.410	—
	masking — ERM	+0.254 [+0.230, +0.278]	−0.083 [−0.112, −0.057]	+0.337 [+0.312, +0.364]
	oracle inpainting — ERM	+0.225 [+0.204, +0.247]	+0.442 [+0.413, +0.470]	−0.216 [−0.235, −0.198]
	group-balanced — ERM	+0.229 [+0.206, +0.251]	+0.442 [+0.414, +0.470]	−0.214 [−0.231, −0.196]
	DFR — ERM	+0.218 [+0.197, +0.240]	+0.433 [+0.406, +0.460]	−0.214 [−0.234, −0.196]
	paired LEACE — ERM	+0.233 [+0.212, +0.255]	+0.452 [+0.424, +0.481]	−0.219 [−0.238, −0.202]

5.4 Sensitivity to the interval estimator

As in Stage 1, both estimators were computed on the same predictions of the other-modality sweeps (Table 10): the crossed intervals are wider by a median factor of 1.45; 11 of 180 contrasts lose significance and none gains it (Table 11). Every statement in this report uses the crossed intervals, so these changes are already reflected in the text: in particular, MedSigLIP’s small benefit at full thyroid overlap and the ovarian harm at 50% overlap are not claimed.

Table 10: The sweeps of the other modalities under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ratio (crossed / per-seed)
synthetic/capsule	60	9		0		1.72
synthetic/nih_ptx	30	0		0		1.44
synthetic/ovary	30	1		0		1.59
synthetic/thyroid	60	1		0		1.37
all	180	11		0		1.45

Table 11: Contrasts of the other-modality sweeps whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
synthetic/thyroid/ medsiglip448_caliper_corr_main	1.0 / mask / erm / test_rev	+0.044 [−0.011, +0.104]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.0 / inpaint / erm / test_rev	+0.024 [−0.016, +0.069]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.75 / inpaint / erm / test_rev	+0.036 [−0.005, +0.074]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	1.0 / inpaint / erm / test_rev	+0.026 [−0.020, +0.074]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.5 / balanced / erm / test_rev	+0.030 [−0.018, +0.076]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.75 / balanced / erm / test_rev	+0.028 [−0.023, +0.076]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.0 / dfr / erm / test_rev	+0.030 [−0.034, +0.090]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.5 / dfr / erm / test_rev	+0.048 [−0.014, +0.110]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	0.75 / dfr / erm / test_rev	+0.055 [−0.010, +0.122]	no longer excludes 0
synthetic/capsule/ dino518_debris_corr_main	1.0 / dfr / erm / test_rev	+0.049 [−0.016, +0.114]	no longer excludes 0
synthetic/ovary/ dino518_caliper_corr_main	0.5 / mask / erm / test_rev	−0.107 [−0.269, +0.054]	no longer excludes 0

6 What failed or changed

- **Harm at full overlap is not universal.** The robust statement is the *interaction* (10 of 11); harm at full overlap is significant in 7 of 11 sweeps.
- **Two lesion-size cells are not significant** (large lesions at full overlap, medium lesions at 50%): harm is ordered by lesion size, not monotone in every cell.
- **Chest radiography with RAD-DINO contradicts the law**, as a failure of its premise. It was registered as a confirmation and is reported as a boundary case.
- **The counterfactual mechanism holds in 3 of 5 sweeps:** not for thyroid calipers (equal sensitivity) and reversed for the chest tube with RAD-DINO.
- **The ovarian sweep has no significant benefit outside the ROI**, so its interaction rests on the harm side.

- **Reproducibility of the capsule split.** The capsule sweep’s split is produced by a seeded shuffle of the group list; rebuilding the cohort (a re-derived contamination mask moved a few frames across the 5% threshold) re-dealt the split, and only 32 of 71 artifact-bearing test frames of an earlier build recur. Point estimates of that sweep moved by up to about 0.2, while the location interaction kept its sign and significance. The split is now frozen with the cohort file.

7 Anticipated questions

Why not simply show saliency maps? They show where a model looks, not how much its decision depends on the artifact, and they are unstable for frozen encoders with linear heads. The same-head counterfactual answers the causal question directly on the same image.

If masking removes context, why is the occlusion control not harmed? Because in the occlusion control the ruler is uninformative: the head does not need it, so keeping it costs nothing, and the context lost by masking is small for this task. The harm in the sweep appears only when the retained ruler predicts the label in training.

Why does dilation make things worse at 50% overlap? Because it makes more of the ruler visible (retention $0.50 \rightarrow 0.95$) while adding only a thin rim of context. The shortcut becomes stronger; the head uses it more.

Why does the harm not grow at 100% overlap when the mask is dilated? The ruler is already fully visible; dilation only adds skin. If harm were about the ruler’s share of the image it would fall; it does not.

Why do the artifact-targeting arms help *more* when the artifact is inside? ERM relies more on a ruler on the lesion (its reversed AUROC is lower at $r = 1$), so removing that reliance gains more. Their gain does not depend on the ROI.

Why does thyroid show loss of benefit rather than harm? The ultrasound background contains little disease information, so masking loses almost nothing besides failing to remove the caliper; ERM is already caliper-driven, and masking leaves it so. Harm needs masking to remove disease evidence as well.

Why is the chest tube a boundary and not a failure? The law compares what masking removes outside and inside the ROI. RAD-DINO does not use the out-of-lung tube at all (reversed $\approx$ clean AUROC), so there is nothing to remove outside and the comparison has no content. DINOv2 uses the tube and shows the law.

Are 3 seeds enough? For the interactions, yes: the intervals include the variation of the fixed test set, and 10 of 11 exclude zero. Individual cells in small cohorts (ovary) are imprecise and are interpreted only through the interaction.

Could the synthetic artifacts be unrealistic? Yes, and that is their purpose: they isolate location. Whether the law holds for artifacts that clinicians actually produce is a separate question that synthetic sweeps cannot answer.

8 Conclusion

Supported. The location law in its interaction form (Q2a) holds in 10 of 11 controlled sweeps across five modalities and four encoders. The mechanism (Q2b) is retention with amplified reliance: occlusion alone does not harm, harm follows the visibility of the artifact under dilation, and the masked head’s same-image reliance on an in-ROI artifact is about 2.1 times that of ERM for the ruler (+0.255 [+0.221, +0.287]) and significantly higher in 3 of 5 sweeps. **Qualified.** Harm at full overlap is setting-dependent; the lesion-size gradient is an ordering. **Not supported.** The law for an encoder that ignores the out-of-ROI artifact (chest tube, RAD-DINO). The practical reading is: masking removes what lies outside the ROI and concentrates the model on what lies inside; whether that helps depends on where the shortcut sits.

9 Questions for the supervisor

1. Should the thesis state the law in its interaction form only (robust) and treat harm at full overlap as setting-dependent, or keep harm as the headline for dermoscopy, where it holds with every encoder?
2. The counterfactual mechanism holds in 3 of 5 sweeps. Is the thyroid result (the unmasked model is already as reliant on an in-nodule caliper as the masked one) best presented as a limitation or as a second route to the same outcome?
3. Is it acceptable to present chest radiography with RAD-DINO as a boundary of the law rather than as a failed confirmation, given that it was registered as a confirmation?

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